



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

– Hira Kumar

Computer Networking

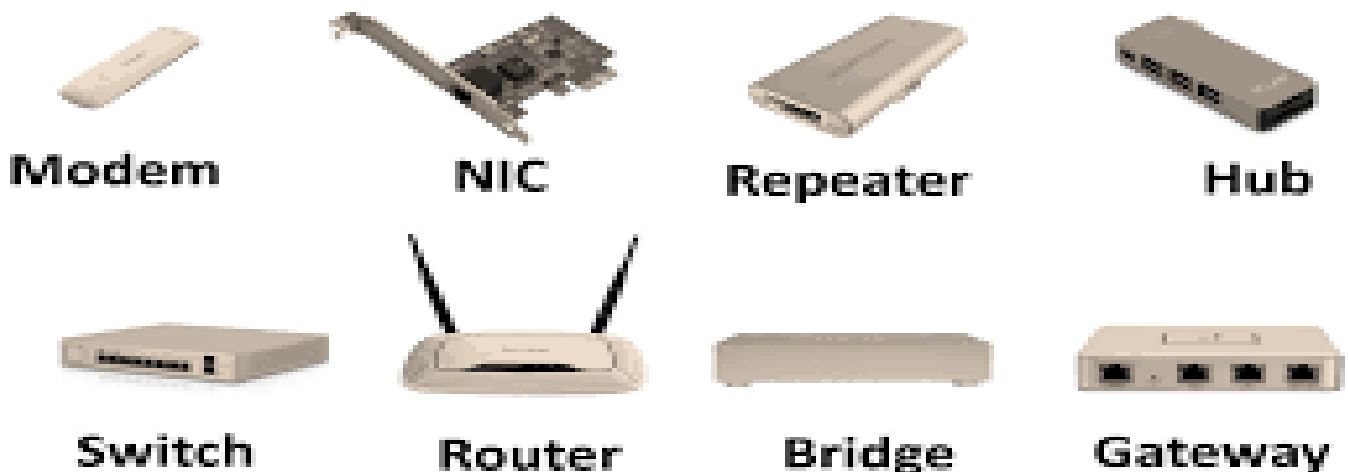


❖ Computer Network Devices

- A computer network device is a physical piece of hardware that allows electronic devices / node / computer to connect, communicate, and share resources on a network.
- These devices move data from one point to another, protect networks, and ensure that data reaches its correct destination.
- Different network devices serve distinct roles depending on how data needs to be directed, filtered, or transmitted.

Various Device in Computer Network

1. NIC - For Network create
 2. Repeaters
 3. Hubs
 4. Bridge
 5. Switch
 6. Router
 7. Access point
 8. Gateway
 9. IDS
 10. Firewall
 11. Modem
- pure h/w device
- H/w & S/w both
- Network security



Types of Network Devices

1. NIC (network interface card) OR NAC (network adapter card)

- NIC is a hardware component/device that allows a computer or

other device to connect to a network and send/receive data.

- NIC can support different data rates such as 10 Mbps, 100 Mbps, 1 Gbps (1000 Mbps), depending on the NIC.
- Every NIC has a unique MAC address, e.g. 9C-45-3F-26-34-07.
- A standard MAC address is 48 bits = 6 bytes.
- MAC address is also called a Physical Address or Hardware Address.
- IEEE defines/standardizes the MAC address format and assigns the manufacturer-specific OUI (first 24 bits); the manufacturer assigns the remaining portion to make the address unique.
- MAC address is normally stored in the NIC's non-volatile memory/firmware.

There are two types of NIC

1. Internal NIC

2. External NIC



➤ Internal NIC

- Installed inside the computer/device.
- Can be wired or wireless.
- Wired NIC (Ethernet NIC): Uses a network cable, commonly with an RJ-45 connector.
- Wireless NIC (Wi-Fi adapter): Uses radio signals and generally has an antenna.

Internal NIC can be either built into the motherboard or installed

separately inside an expansion slot.

Internal NIC – 2 forms

1. Onboard/Built-in NIC

- Already integrated into the motherboard.
- No separate card is required.
- Example: Most modern desktop motherboards have an integrated Ethernet NIC.

2. Expansion/Separate NIC

- A separate NIC card is installed in an expansion slot, commonly PCIe.
- Example: PCIe Ethernet card, PCIe Wi-Fi card.

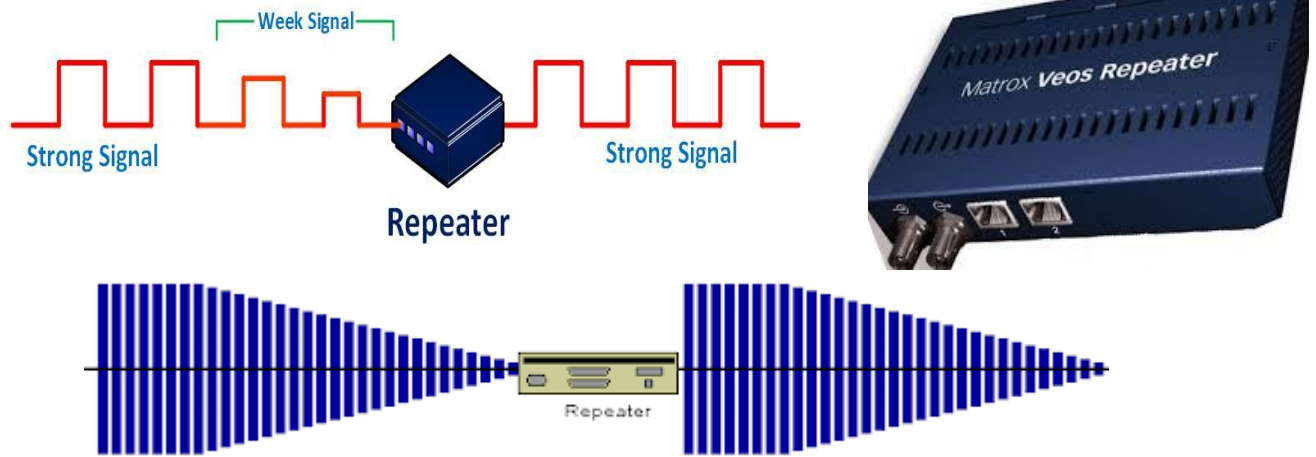
➤ **External NIC**

- Connected externally through a port such as **USB**.
- Examples: **USB Wi-Fi adapter, USB Ethernet adapter/dongle, bluetooth dongle.**

2. Repeater

- **Definition:** A repeater is an electronic hardware device that receives a weak or degraded network signal, regenerates (amplifies) it, and retransmits it at its original strength.
- **Core Function:** Its primary purpose is to extend the physical coverage area of a network beyond the normal distance limits of cables or wireless signals (e.g., Ethernet cables lose signal strength after 100 meters).
- **OSI Layer:** It works on the Physical Layer of the OSI model because it only deals with electrical, optical, or radio signals.
- **Intelligence:** It is a non-intelligent device. It does not look at MAC addresses, IP addresses, or data content; it simply copies the incoming signal and sends it out stronger.
- **Limitations:** A repeater does not filter traffic or fix data errors. If it receives a corrupted signal or background noise, it will amplify the noise along with the data.
- **Analogy & Use:** Analogy: A megaphone or loudspeaker. It doesn't change what you are saying; it just makes your voice loud enough to reach people standing far away. Commonly used in long-distance

fiber-optic cables, large office buildings, and home Wi-Fi extenders.



3. HUB

- **Definition:** A hub is a hardware device used to connect multiple devices on a network and share the network connection among them.
- **Collision Limitation:** A major limitation of a hub is that if two devices send data at the exact same time, a data collision occurs.
- **Transmission Mode:** A hub operates in half-duplex mode, meaning devices can send and receive data, but not at the same time (it is not strictly simplex, as two-way communication is possible, just not simultaneous).
- **OSI Layer & Intelligence:** It works on the Physical Layer of the OSI model. It is a non-intelligent device because it broadcasts data to every connected port, making it insecure and inefficient.
- **Analogy & Use:** Analogy: A WhatsApp group where anyone can send a message, but everyone is forced to see it. It was commonly used in legacy home networks but is rarely used today.



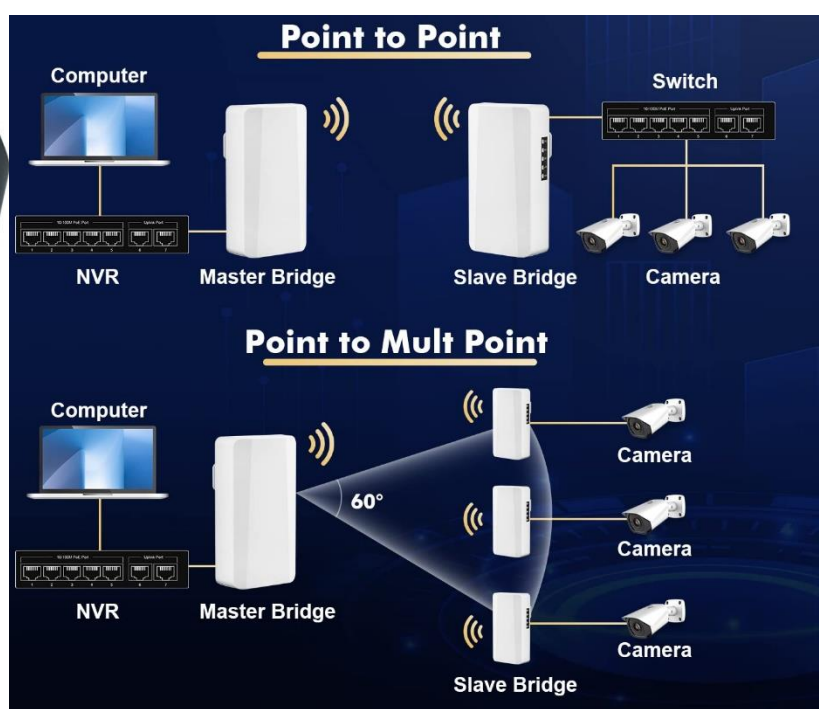
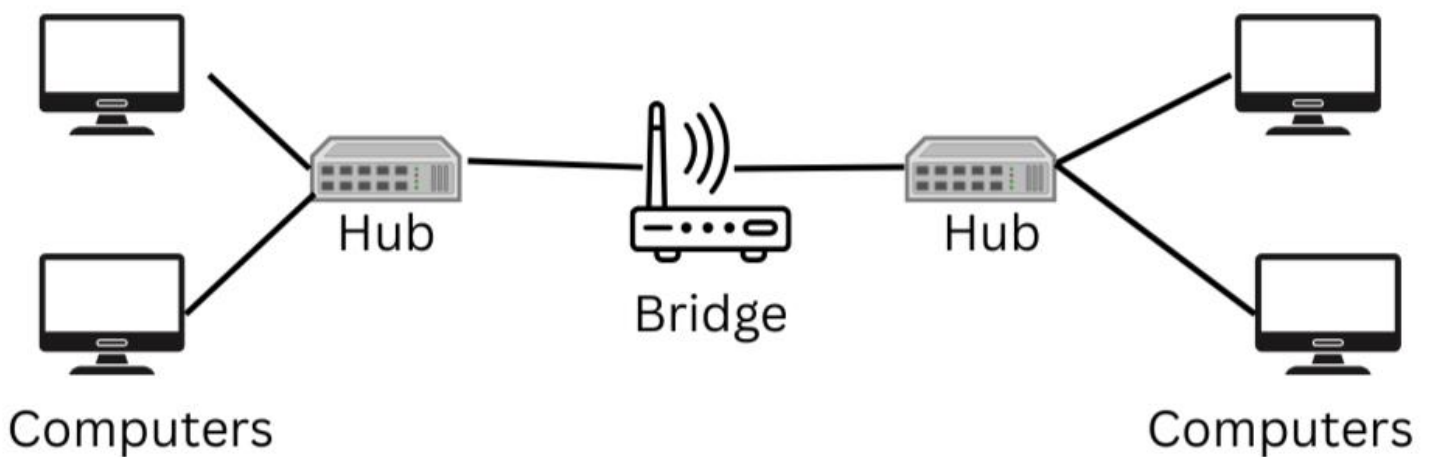
4. Bridge

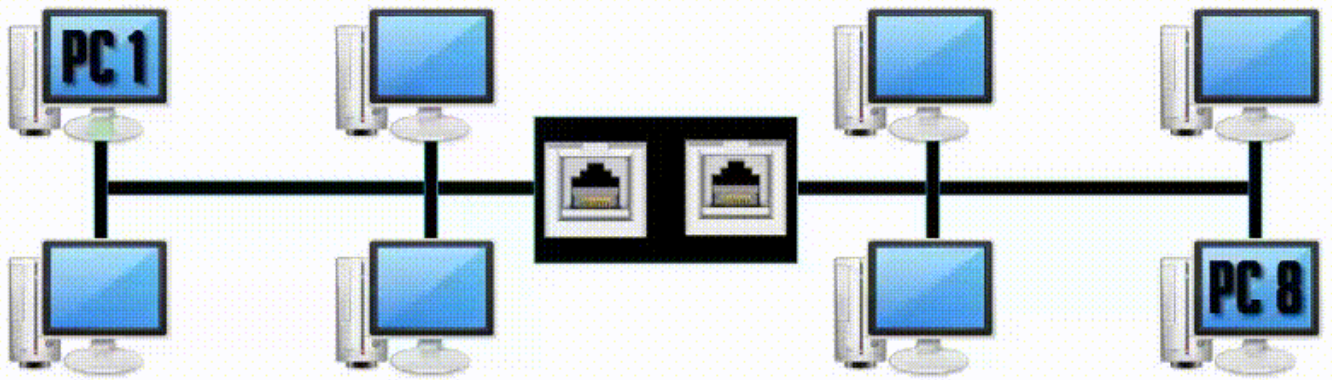
- **Definition:** A bridge is a hardware device used to connect two separate Local Area Networks (LANs) or network segments into a single, unified network.
- **OSI Layer:** It works on the Data Link Layer of the OSI model.
- **Filter and Forward:** A bridge inspects incoming traffic and decides whether to filter (block) it or forward it across the bridge based on the destination MAC address. This reduces unnecessary network traffic.
- **Ports:** It typically has only two ports (one for each network segment), making it different from a switch, which has many ports.
- **Intelligence:** It is an intelligent device because it maintains a MAC address table to learn which devices belong to which side of the

network.

- **Analogy & Use:** Analogy: A physical bridge connecting two islands; it only lets traffic cross if the destination is on the other side. Commonly used to split a large, congested network into two smaller, quieter sections.
- **Types :** 2 type (i) Static (ii) Dynamic/ transparent

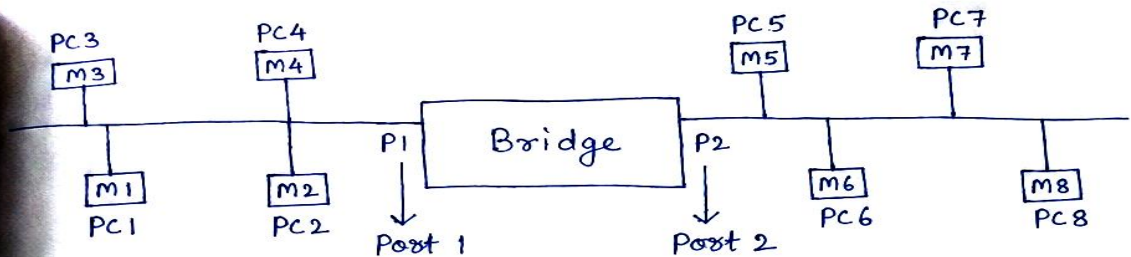
Feature	Static Bridge	Dynamic Bridge
MAC Address Table	Configured manually	Learned automatically
Learning	Does not learn MAC addresses automatically	Learns MAC addresses automatically
Updating	Administrator must update it	Updates automatically
Usage	Less common	Common in modern networks





SEGMENT 1

SEGMENT 2



Frame Contains 2 MAC Address

Source	Destination
--------	-------------

M → MAC

PC → Personal Computer

Static Bridges Table

MAC	PORT
M1	P1
M2	P1
M3	P1
M4	P1
M5	P2
M6	P2
M7	P2
M8	P2

Dynamic Bridges Table

MAC	PORT
M1	P1
M2	P1
M3	P1
M4	P1
M5	P2
M6	P2
M7	P2
M8	P2

* Dynamic MAC table me bridge source MAC address ko dekhkar ye seekhta hai ki kaun-sa MAC address kis port par maujood hai.

❖ What is segment

- A segment is a portion/part of a network.
- For example, if a bridge divides one LAN into two parts:

PC1 — PC2 — Bridge — PC3 — PC4

Then:

PC1–PC2 = one network segment

PC3–PC4 = another network segment

❖ What is flood

- Flooding means sending a frame to all ports/devices in the same network segment, except the port from which the frame was received.
- For example, a switch receives a frame but doesn't know the destination MAC address:

PC1 → Switch → PC2, PC3, PC4

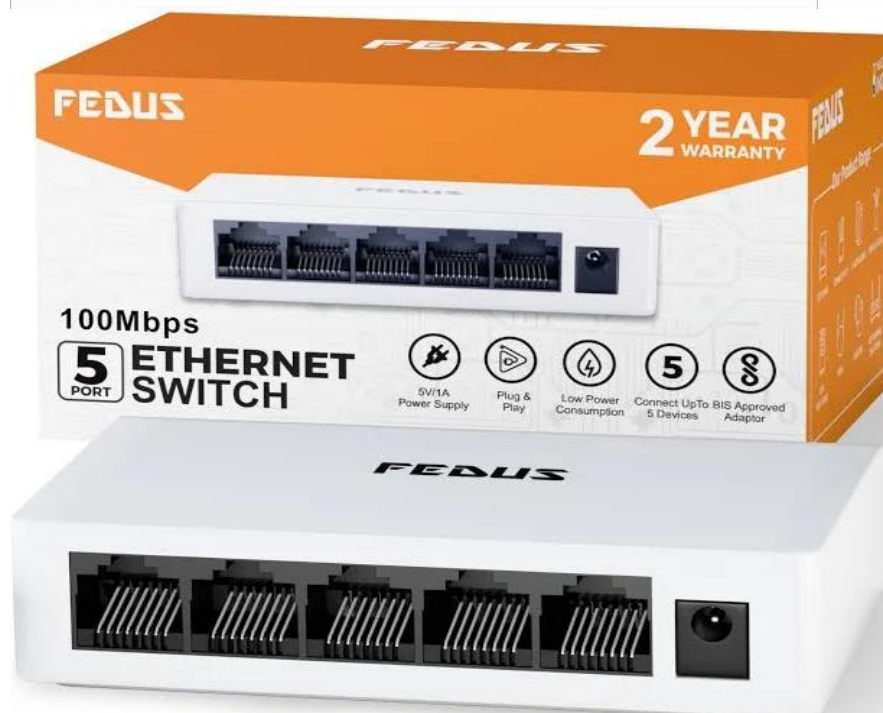
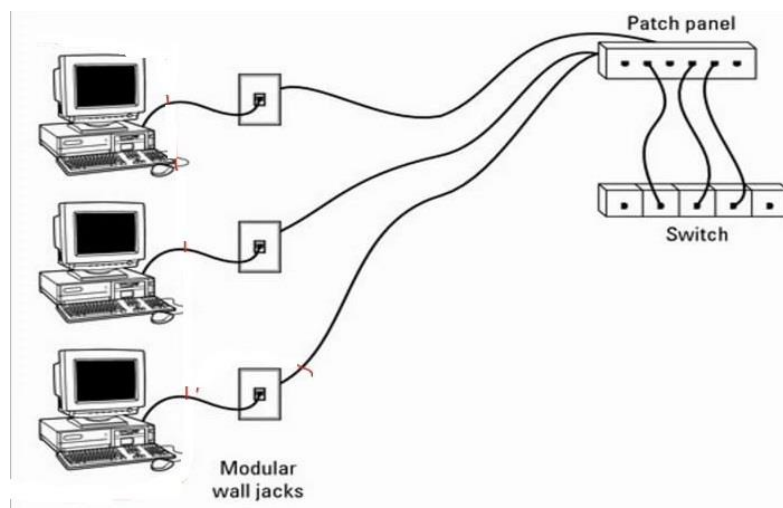
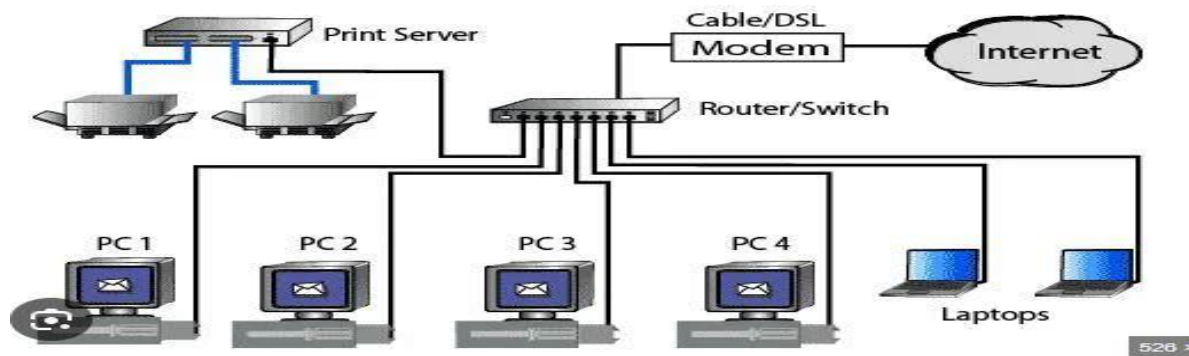
The switch forwards the frame to PC2, PC3 and PC4.

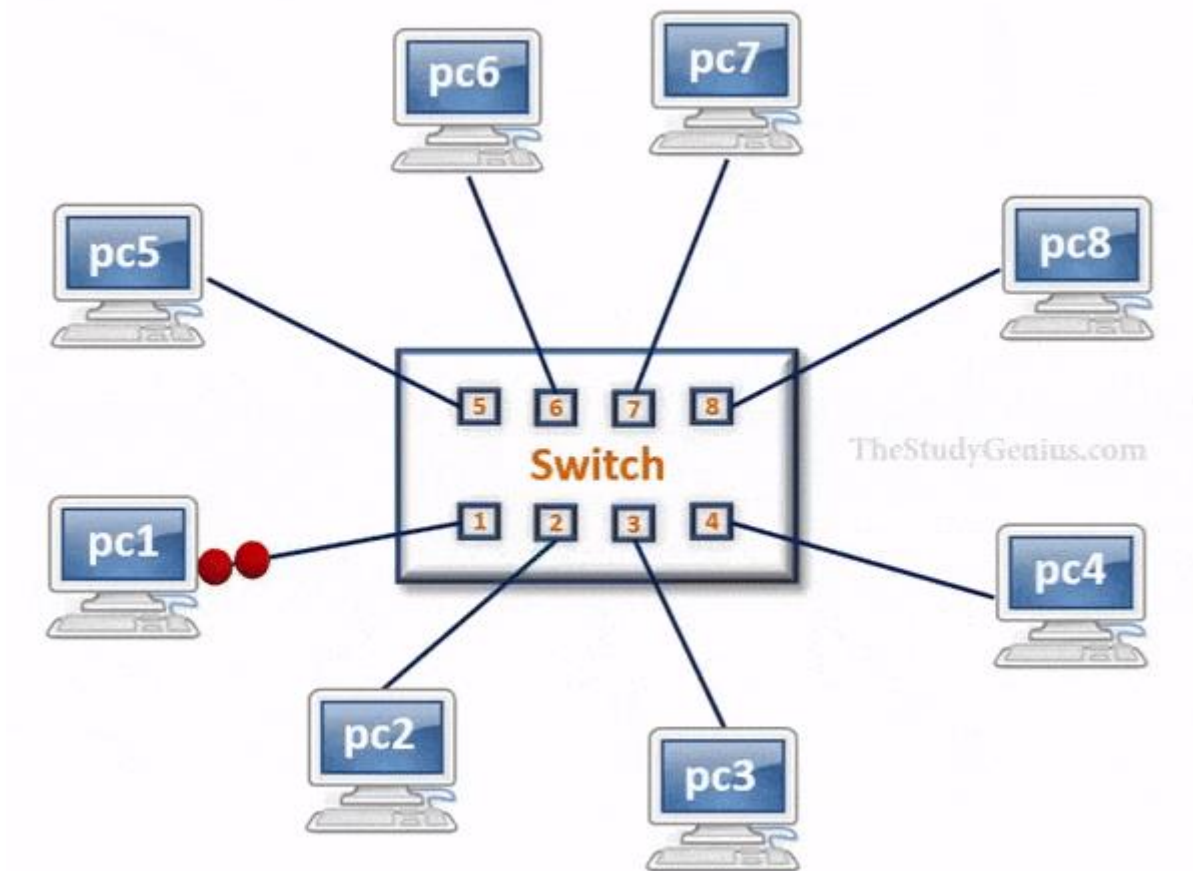
This is called flooding.

5. Switch

- **OSI Layer:** It works on the Data Link Layer of the OSI model.
- **Multiport Bridge:** It is often referred to as a multiport bridge because it connects multiple segments of a network.
- **Transmission Mode:** It operates in full-duplex mode, allowing devices to send and receive data simultaneously without collisions. It is an intelligent device.
- **Filter and Forward:** A switch performs data filtering and forwarding by reading the destination MAC address.
- **Definition:** A switch is a hardware device that connects multiple devices within a local computer network (LAN).
- **Advanced Features:** It is much more advanced than a hub. A switch maintains a dynamic MAC address table (**MAC address learning**) to decide exactly where data should be transmitted. It delivers the message only to the specific physical address (MAC address) specified in the incoming data packet.

- **Data Flow:** In a network, if PC1 wants to send data, the switch can send it directly to a specific device (like PC2, PC3, PC4, a laptop, or a printer) rather than broadcasting it to everyone.
- **Common Uses:** Commonly used in high-traffic environments like banks, offices, and schools.





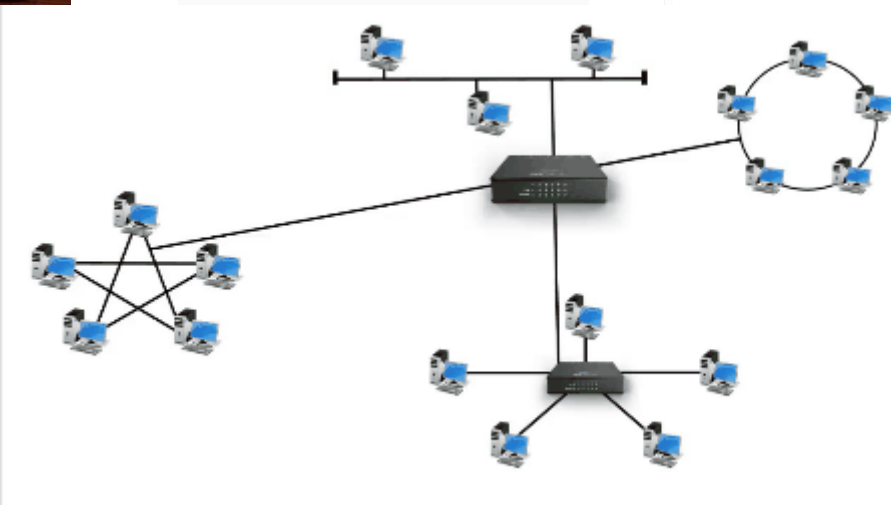
Types of Switches

5 types of switches

1. **Unmanaged Switches:** Simple plug-and-play devices with no setup needed. They are best for home networks or small offices.
2. **Managed Switches:** Advanced devices that allow traffic monitoring, security rules, and network customization. They are used in large businesses.
3. **Smart Switches / Layer2 switch:** A middle option between unmanaged and managed types. They offer a web page for basic settings like VLANs.
4. **PoE (Power over Ethernet) Switches:** Send both data and electrical power through one Ethernet cable to power IP cameras or phones.
5. **Layer 3 Switches:** Combine the normal job of a switch with the routing power of a router.

6. Router

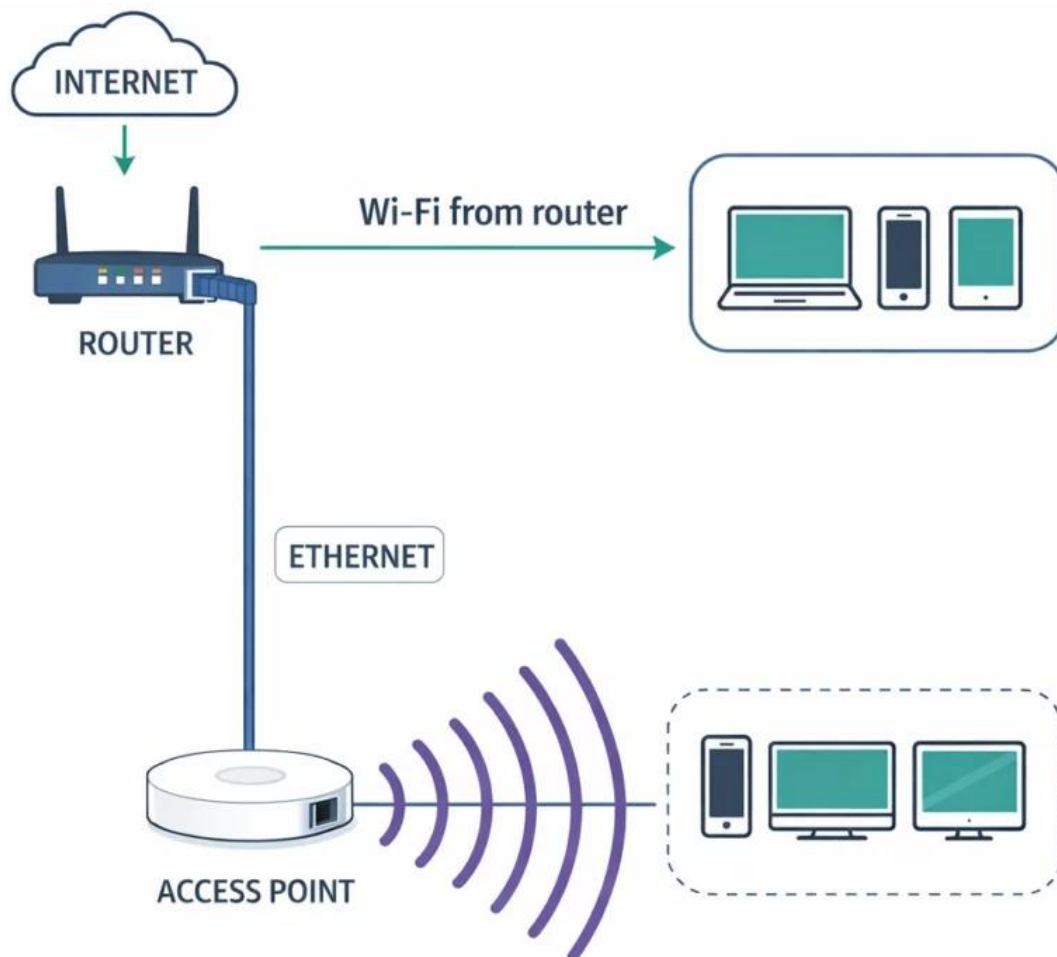
- **Definition:** A router is a highly intelligent hardware device used to connect entirely different networks together, such as connecting your private home LAN to the public internet.
- **OSI Layer:** It works on the Network Layer of the OSI model.
- **Routing Table & Best Path:** A router maintains a routing table and uses routing protocols to determine the absolute shortest and best path for data packets to travel across networks.
- **Addressing:** Unlike switches and bridges that use physical MAC addresses, a router uses logical IP addresses to transmit data.
- **Transmission Mode:** It operates in full-duplex mode and acts as a security barrier by utilizing built-in firewall features and NAT (Network Address Translation).
- **Analogy & Use:** Analogy: A postal sorting office that looks at the city and zip code (IP address) on a letter to route it to the correct destination country or state. Commonly used in homes, offices, banks, and internet service providers (ISPs) to provide internet access.



HUB	SWITCH	ROUTER
HUB works in Physical Layer of OSI Model.	SWITCH works in Data Link Layer of OSI Model.	ROUTER works in Network Layer of OSI Model
HUB is a broadcast device	SWITCH is a multicast device	ROUTER is a routing device use to create route for transmitting data packets.
HUB is use to connect devices in the same network.	SWITCH is use to connect devices in the same network	ROUTER is use to connect two or more different network.
HUB sends data in the form of packets.	SWITCH sends data in the form of frames.	ROUTER sends data in the form packets.
HUB only works in half duplex.	SWITCH works in full duplex.	ROUTER works in full duplex.
Only one device can send data at a time.	Multiple devices can send data at the same time.	Multiple devices can send data at the same time.
HUB does not store any mac/ IP address to transfer data.	SWITCH stores and uses MAC address of a devices to transfer data.	ROUTER uses IP address to transfer data.

7. Access Point (Wireless Access Point / WAP)

- **Definition:** An Access Point is a hardware device that adds wireless capability to a wired network by creating a Wireless Local Area Network (WLAN).
- **Core Function:** It plugs directly into a wired router or switch via an Ethernet cable and projects a Wi-Fi signal, allowing wireless devices (like smartphones, laptops, and smart TVs) to connect to the network.
- **OSI Layer:** It primary works on the Data Link Layer and Physical Layer of the OSI model.
- **Range Extension:** It is commonly used to extend the wireless coverage of an existing network and support a large number of simultaneous wireless connections.
- **Intelligence:** It is a semi-intelligent bridge device. It converts wired Ethernet data into wireless radio signals (and vice versa) and enforces wireless security protocols (like WPA2/WPA3) using password authentication.
- **Difference from a Router:** A router assigns IP addresses and connects different networks (like your home to the internet). An Access Point cannot assign IP addresses; it simply extends the router's network wirelessly.
- **Analogy & Use:** Analogy: An extension cord for electricity. It doesn't generate the power (internet), but it extends it to a place where you couldn't reach before. Commonly used in large offices, universities, hotels, and big homes where a single Wi-Fi router cannot cover the entire building.

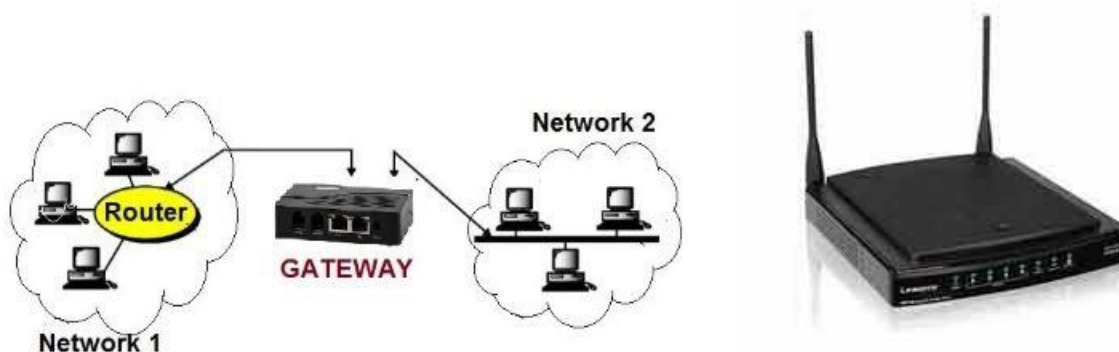


8. Gateways

- **Definition:** A gateway is a specialized network device (or software) that acts as a protocol converter, allowing two entirely different networks with different architectures, protocols, and data formats to communicate with each other.
- **Core Function:** It acts as the "entry and exit point" for a network. It takes data from one network, translates it into a language or format that the destination network can understand, and then sends it through.
- **OSI Layer:** It can work on any layer of the OSI model, but it most

commonly operates at the Application Layer (Layer 7) or Network Layer (Layer 3) depending on the type of translation required.

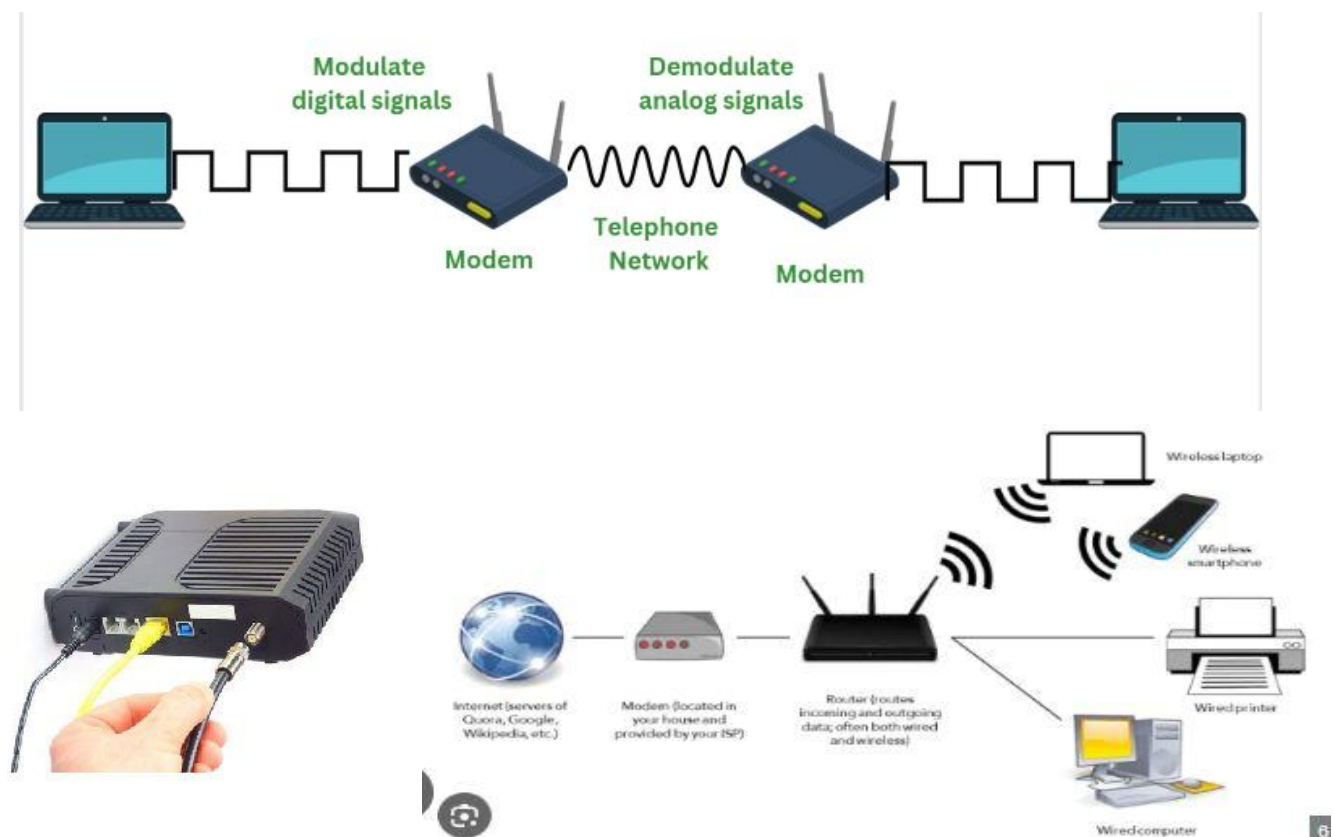
- **Intelligence:** It is a highly intelligent device because it performs deep packet inspection, data translation, encryption, and protocol conversion rather than just forwarding data.
- **Difference from a Router:** A router connects similar networks using the same protocol (like connecting two IP networks). A gateway connects completely dissimilar networks (LAN, WAN, PAN, MAN,) using different protocols (like connecting an old banking mainframe network to the modern internet). *Note: In home networks, your router acts as a default gateway because it translates your private home network traffic into public internet traffic.*
- **Analogy & Use:** Analogy: A human language translator standing between an English speaker and a Hindi speaker. Without the translator, they cannot understand each other. Commonly used in enterprise businesses, cloud computing (e.g., connecting a local office to AWS), and e-commerce payment processing, ISP.



9. Modem

- **Definition:** A modem is a hardware device that converts digital data from a computer into an analog signal for transmission over telephone or cable lines, and vice versa.
- **Core Function:** It performs two main tasks: **Modulation** (converting digital signals from your devices into analog signals to send over the provider's lines) and **Demodulation** (converting incoming analog signals from the provider back into digital signals your devices can understand).
- **OSI Layer:** It primarily works on the Physical Layer and Data Link Layer of the OSI model.

- **Bridge to the Internet:** It serves as your essential bridge to your Internet Service Provider (ISP). Without a modem, your home network cannot connect to the external internet infrastructure.
- **Difference from a Router:** A modem brings the internet connection into your home from the outside world. A router takes that single internet connection from the modem and shares it among your multiple devices (phones, laptops, TVs).
- **Analogy & Use:** Analogy: A translator between someone who only speaks sign language (digital) and someone who only speaks spoken words (analog). Commonly used in every home and office that has an internet connection (such as DSL, cable, or fiber-optic broadband).



10. Dongle

- **Definition:** A network dongle is a small, pocket-sized USB device that plugs into a laptop or desktop computer to give it instant network or internet access.
- **Core Function:** It acts as an external wireless adapter / External NIC. It captures external signals (like Wi-Fi or cellular data) and translates them so your computer can connect to a network without needing a physical Ethernet cable.

- **OSI Layer:** It operates primarily on the **Physical Layer** and **Data Link Layer** of the OSI model, as it sends and receives raw data signals.
- **Types of Network Dongles:**
 - ✓ **Wi-Fi Dongle:** Adds Wi-Fi capabilities to an old desktop or laptop that doesn't have a built-in Wi-Fi card.
 - ✓ **Cellular Dongle (4G/5G Internet Dongle):** Contains a slot for a SIM card. It connects your computer directly to mobile network towers, giving you internet access anywhere, even without Wi-Fi.
- **Intelligence:** It is a semi-intelligent, plug-and-play device. It contains its own small internal modem and firmware to safely process network data.
- **Analogy & Use:** Analogy: A portable key or pass that unlocks wireless features. Commonly used by travelers, remote workers, or people with older computers who need quick, mobile internet access on the go.

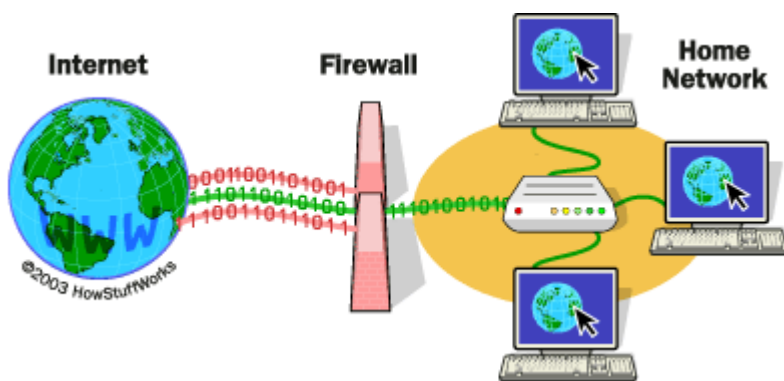


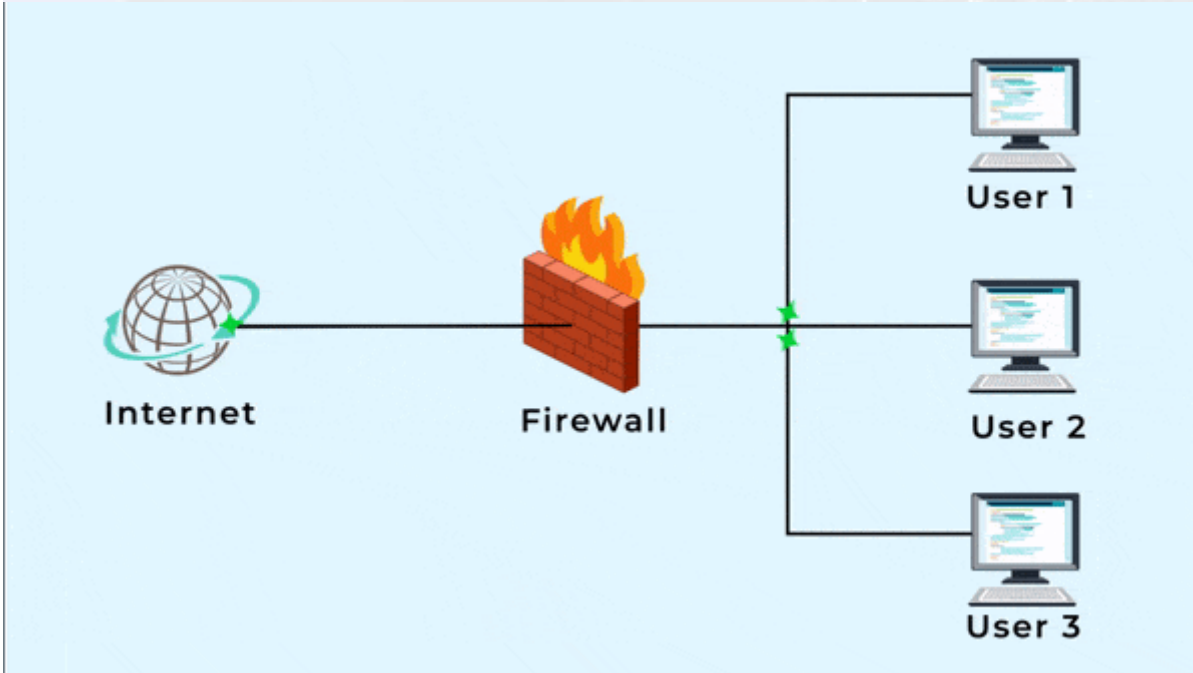
11. Firewall

- **Definition:** A firewall is a network security device (or software application) that monitors and filters incoming and outgoing network traffic based on an organization's previously established security policies.
- **Core Function:** It acts as a protective barrier or "shield" between a trusted internal network (like your private office LAN) and an untrusted external network (like the public internet) to block unauthorized access and cyberattacks.
- **OSI Layer:** It can work on multiple layers depending on its type. Traditional firewalls work at the **Network Layer** (Layer 3) and **Transport Layer** (Layer 4) to filter IP addresses and ports, while advanced Next-Generation Firewalls (NGFW) work up to the **Application Layer** (Layer 7) to inspect actual data content.

- **Types:**

- **Hardware Firewall:** A physical device placed between the router and the internet to protect the entire network.
- **Software Firewall:** A program installed directly on an individual computer (like Windows Defender) to protect just that single machine.
- **Intelligence:** It is a highly intelligent, proactive device. It uses advanced rulesets, packet inspection, and access control lists (ACLs) to analyze traffic and automatically decide whether to **Allow (Pass)** or **Block (Drop)** data.
- **Analogy & Use:** Analogy: A security guard standing at the main gate of a private building. The guard checks everyone's ID and bag against a guest list before letting them inside. Commonly used in **all banks, corporate offices, schools, and personal computers** to prevent hacking, malware, and data theft.





12. IDS (Intrusion Detection System)

- **Definition:** An IDS is a network security device or software application that monitors network traffic for suspicious activities, known threats, or policy violations.
- **Core Function:** It acts as a passive surveillance system. It analyzes traffic patterns and compares them against a database of known cyberattacks or unusual behaviors. If it detects a threat, it instantly generates an **alert** to notify network administrators.
- **OSI Layer:** It operates across multiple layers, primarily at the **Network Layer** (Layer 3), **Transport Layer** (Layer 4), and **Application**

Layer (Layer 7) to inspect packet headers and data payloads.

- **Types:**

- **NIDS (Network-based IDS):** Plugs into strategic points of a network to monitor traffic going to and from *all* devices.
- **HIDS (Host-based IDS):** Installed directly on a specific, critical device (like a server) to monitor only that single machine's activity.

- **Intelligence:** It is a highly intelligent, diagnostic system. It uses signature-based detection (matching known virus signatures) and anomaly-based detection (finding unusual spikes in traffic) to identify hackers or malware.
- **Difference from a Firewall & IPS:** A **Firewall** acts as a barrier to block unauthorized entry. An **IDS** sits *inside* the network to watch for anything sneaky that successfully bypassed the firewall. Unlike an **IPS (Intrusion Prevention System)** which automatically blocks a threat, a standard **IDS** only *detects and alerts*—it does not stop the attack on its own.
- **Analogy & Use:** Analogy: A security camera or a smoke alarm inside a building. It cannot put out the fire or tackle a robber, but it loudly rings an alarm to let you know there is trouble. Commonly used alongside firewalls in **banks, cloud data centers, and corporate IT infrastructures** to catch hidden network threats.

=====HIRA KUMAR=====